(i) F has a partial derivative with respect to x at $(x_0, y_0) \in A$ if $\lim_{h \to 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h}$ exists and it is denoted by $\left.\frac{\partial F}{\partial x}\right|_{(x_0, y_0)}$.

F has a partial derivative with respect to y at $(x_0, y_0) \in A$ if $\lim_{k \to 0} \frac{F(x_0, y_0 + k) - F(x_0, y_0)}{k}$ exists and limit value is defined by $\left.\frac{\partial F}{\partial y}\right|_{(x_0, y_0)}$.

- Clariant's Theorem: Suppose that $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}$. If f_{xy} and f_{yx} exist in A and are continuous in A, then $f_{xy} = f_{yx}$ in A.
- Let $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$. A function $U : A \to \mathbb{R}$ is said to be harmonic in A if it satisfies $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \forall (x, y) \in A$. This equation is called Laplace's equation.
- Let $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}$ and $(x_0, y_0) \in A$.

  (i) The linear approximation of F at $(x_0, y_0) \in A$ is defined to be

  $$L(x, y) = F(x_0, y_0) + \left.\frac{\partial F}{\partial x}\right|_{(x_0, y_0)} (x - x_0) + \left.\frac{\partial F}{\partial y}\right|_{(x_0, y_0)} (y - y_0)$$

  (ii) The differential of F is defined to be $dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$ where $\Delta x = dx$ and $\Delta y = dy$.
- Suppose w is a function of two variables x, y where x and y are functions of a single variable 't' then $\frac{dw}{dt} = \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt}$
- Suppose w is a function of two variables x and y where x and y are functions of two variables s and t then, $\frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial s}$, $\frac{\partial w}{\partial t} = \frac{\partial w}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial w}{\partial y} \cdot \frac{\partial y}{\partial t}$
- Suppose that $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}^2$. If F is having continuous partial derivatives and homogeneous on A, with degree p, then $x \frac{\partial F}{\partial x} + y \frac{\partial F}{\partial y} = pF$.

ICT CORNER

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